Notes Short Horizon

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23 juillet 2026

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1 Practitioner Regret

1.1 Practitioner Regret and the Definition of $\varepsilon_\eta(T, \nu)$

We revisit the definition of the practitioner regret introduced in *Short Horizon Reinforcement Learning.*

1.1.1 Environment and Notation

Consider a stochastic K-armed bandit environment, denoted by ν. At each time step, an agent selects an action from a finite set of arms $\mathcal{A} = \{1, 2, \ldots, K\}$. Each arm $a \in \mathcal{A}$ is characterized by an unknown probability distribution with expected reward μ_a. We denote the optimal expected reward in the environment as :

$$\mu^\star = \max_{a \in \mathcal{A}} \mu_a \tag{1.1}$$

and we define the standard sub-optimality gap for any given arm a as $\Delta_a = \mu^\star - \mu_a$.

To accommodate the notion of practitioner regret, we introduce a relaxed performance baseline parameterized by a precision level $\varepsilon \geq 0$. We define the set of *ε-near-optimal arms* as those whose expected rewards are within ε of the true optimum :

$$\mathcal{O}_\varepsilon = \{a \in \mathcal{A} : \mu_a \geq \mu^\star - \varepsilon\} \tag{1.2}$$

For arms outside this near-optimal set, we measure their performance drop relative to the relaxed baseline rather than the absolute true optimum. Thus, the *ε-optimality gap* of an arm a is defined as :

$$\Delta_\varepsilon(a) = (\mu^\star - \varepsilon - \mu_a)_+ \tag{1.3}$$

where $(x)_+ = \max(0, x)$. By definition, for any arm $a \in \mathcal{O}_\varepsilon$, the relaxed gap $\Delta_\varepsilon(a)$ is exactly zero.

Finally, let $N_T(a)$ denote the number of times arm a is pulled up to a time horizon T. The *ε-optimal regret* of an algorithm A interacting with environment ν is the cumulative expected loss incurred exclusively by pulling strictly sub-optimal arms at level ε :

$$R_T^\varepsilon(A, \nu) = \sum_{a \notin \mathcal{O}_\varepsilon} \mathbb{E}[N_T(a)] \Delta_\varepsilon(a) \tag{1.4}$$

To quantify when asymptotic regret bounds become informative, the practitioner regret framework introduces a critical horizon. For each arm a, define

$$\tau_\varepsilon(a) = \inf \left\{ T : \frac{1}{K_{\inf}(\nu_a, \mu^\star - \varepsilon)} < \frac{T}{\log T} \right\}.$$

Intuitively, $\tau_\varepsilon(a)$ represents the smallest horizon for which the asymptotic regret contribution of arm a becomes meaningful.

A relaxed version controlled by a parameter $\eta \in (0, 1]$ is defined as

$$\tau_{\varepsilon}^{\eta}(a) = \inf\left\{T : \frac{1}{K_{\text{inf}}(\nu_a, \mu^{\star} - \varepsilon)} < \frac{\eta T}{\log T}\right\}.$$

The precision level used in practitioner regret is obtained by requiring that the horizon exceeds the critical time of every arm.

1.1.2 Definition of $\varepsilon_{\eta}(T, \nu)$

The practitioner precision level is defined as

$$\varepsilon_{\eta}(T, \nu) = \inf\left\{\varepsilon : T \geq \max_{a} \tau_{\varepsilon}^{\eta}(a)\right\}.$$

We search for the smallest precision level ε such that *all arms* are already in the asymptotic regime.Then, the **practitioner regret** is then defined as

$$R_T^{\eta,\dagger}(A, \nu) = R_T^{\varepsilon_{\eta}(T,\nu)}(A, \nu).$$

1.2 Computation of $\epsilon_{\eta}(T, \nu)$ for Gaussian Bandits

As a foundational step before analyzing Bernoulli environments, we first determine the behavior of the precision level $\epsilon_{\eta}(T, \nu)$ under Gaussian distributions. This allows us to exploit the properties of the Gaussian Kullback-Leibler divergence to obtain closed-form analytical conditions, providing a rigorous theoretical backbone for the computational procedures that follow.

Information-Domain Translation of the Precision Level

To substantiate the preceding intuition, we provide a more direct analytical perspective on the theoretical definitions. We begin with the foundational definition of the practitioner's precision level :

$$\epsilon_{\eta}(T, \nu) = \inf\left\{\epsilon \geq 0 : T \geq \max_{a:\Delta_a > \epsilon} \tau_{\eta}^{\epsilon}(a)\right\} \tag{1.5}$$

Let $f(t) = \frac{\log t}{t}$. This function is strictly monotonically decreasing for all $t > e$. Applying a strictly decreasing function to both sides of an inequality inherently reverses its direction :

$$T \geq \tau \iff f(T) \leq f(\tau) \tag{1.6}$$

Furthermore, mapping a maximum operator through a strictly monotonically decreasing function naturally converts it into a minimum operator :

$$f\left(\max_{a} \tau_{\eta}^{\epsilon}(a)\right) = \min_{a} f\left(\tau_{\eta}^{\epsilon}(a)\right) \tag{1.7}$$

By the theoretical definition of the critical time, the instance $\tau_{\eta}^{\epsilon}(a)$ occurs exactly when the temporal uncertainty curve $f(t)$ intersects the static information barrier of the arm :

$$f\left(\tau_{\eta}^{\epsilon}(a)\right) = \eta \cdot \mathcal{K}_{\text{inf}}(\nu_a, \mu^* - \epsilon) \tag{1.8}$$

Substituting Equations 1.7 and 1.8 back into the inequality condition of Equation 1.5, we obtain an isomorphic condition :

$$\frac{\log T}{T} \leq \min_{a:\Delta_a > \epsilon} \left[\eta \cdot \mathcal{K}_{\text{inf}}(\nu_a, \mu^* - \epsilon)\right] \tag{1.9}$$

Consequently, the precision level can be elegantly rewritten as a minimax optimization problem strictly within the information domain :

$$\epsilon_\eta(T,\nu) = \inf\left\{\epsilon \geq 0 : \frac{\log T}{T} \leq \eta \cdot \min_{a:\Delta_a>\epsilon} \mathcal{K}_{\inf}(\nu_a, \mu^* - \epsilon)\right\} \tag{1.10}$$

Gaussian Divergence and the Distinguishability Scale

To operationalize this theoretical definition, we examine its behavior under standard Gaussian reward distributions. Let each arm $a \in \mathcal{A}$ yield rewards governed by a Gaussian distribution with an unknown mean μ_a and a known common variance proxy σ^2. Under this formulation, the Kullback-Leibler divergence to the ϵ-permissible sub-optimal boundary simplifies to the classical quadratic information distance :

$$\mathcal{K}_{\inf}(\nu_a, \mu^* - \epsilon) = \frac{(\Delta_a - \epsilon)^2}{2\sigma^2} \tag{1.11}$$

where $\Delta_a = \mu^* - \mu_a$ denotes the sub-optimality gap of arm a. Substituting this explicit divergence into the information-domain minimax characterization yields the following localized resolution condition for any unpardoned arm $(\Delta_a > \epsilon)$:

$$\frac{\log T}{T} \leq \eta\frac{(\Delta_a - \epsilon)^2}{2\sigma^2} \iff \Delta_a - \epsilon \geq \sqrt{\frac{2\sigma^2 \log T}{\eta T}} \tag{1.12}$$

This structural relation naturally isolates a fundamental sample-dependent threshold, which we define as the *statistical distinguishability scale* :

$$c_T = \sqrt{\frac{2\sigma^2 \log T}{\eta T}} \tag{1.13}$$

The quantity c_T acts as the asymptotic noise floor of the learning process at horizon T. It delineates the sharp geometric boundary of statistical resolution : an arm can be explicitly distinguished from the optimal action if and only if its sub-optimality gap exceeds the combined threshold $\epsilon + c_T$.

Discrete Path Justification for the Iterative Algorithm

The mapping into the Gaussian information domain transforms the abstract infimum problem into a precise geometric search. Let $\mathcal{A}$ be the finite set of arms. For a given precision level ϵ, we define the *indistinguishability band* as the half-open interval $(\epsilon, \epsilon + c_T]$. Any arm whose gap falls within this zone is statistically ambiguous under the current sampling budget. Formally, we collect these arms into the active ambiguous set :

$$\mathcal{A}_\epsilon = \{a \in \mathcal{A} : \epsilon < \Delta_a \leq \epsilon + c_T\} \tag{1.14}$$

The optimization challenge requires finding the global infimum $\epsilon^* = \inf\{\epsilon \geq 0 : \mathcal{A}_\epsilon = \emptyset\}$. Because $\mathcal{A}$ is finite, the constraint set undergoes discrete phase transitions, implying that $\mathcal{A}_\epsilon$ is a piecewise constant step function. This discrete topology justifies the exact mechanics of our iterative relaxation path through a straightforward proof by contradiction.

Suppose the computational procedure initializes at a baseline $\epsilon^{(0)} = 0$ and identifies a non-empty ambiguous set $\mathcal{A}_{\epsilon^{(0)}}$. Let $\Delta_{\max} = \max_{a\in\mathcal{A}_{\epsilon^{(0)}}} \Delta_a$ represent the upper boundary of the active violation. If we attempt to relax the precision level to any intermediate value $\epsilon' \in [\epsilon^{(0)}, \Delta_{\max})$, the bottleneck arm tracking $\Delta_{\max}$ satisfies :

1. $\Delta_{\max} > \epsilon'$, proving that the arm remains unpardoned.
2. $\Delta_{\max} \leq \epsilon^{(0)} + c_T \leq \epsilon' + c_T$, by the definition of the initial band and the monotonicity of ϵ.

Consequently, $\epsilon' < \Delta_{\max} \leq \epsilon' + c_T$, meaning the bottleneck arm remains trapped inside the shifted indistinguishability band $\mathcal{A}_{\epsilon'}$.

This proves that no intermediate continuous adjustment within $[\epsilon^{(0)}, \Delta_{\max})$ can satisfy the global resolution constraint. To minimize the relaxation error while achieving feasibility, the precision level must execute a sharp, discrete jump exactly to the upper boundary, setting $\epsilon^{(1)} = \Delta_{\max}$. This analytical necessity establishes that the sequential evaluation of the boundary $\max \mathcal{A}_\epsilon$ is not a heuristic approximation, but the mathematically rigorous trajectory required to compute the exact infimum, thereby laying the direct groundwork for the following algorithm.

Algorithm 1 Computation of $\epsilon_\eta(T, \nu)$ for Gaussian Bandits

Input: Sub-optimality gaps $\{\Delta_a\}_{a=1}^K$, horizon T, variance σ^2, $\eta \in (0, 1]$
Output: Optimal precision level ϵ^*

- Compute statistical distinguishability scale : $c_T = \sqrt{\frac{2\sigma^2 \log T}{\eta T}}$
- Extract strictly sub-optimal gaps ($\Delta_a > 0$) and sort increasingly :
$\Delta_{(1)} \leq \Delta_{(2)} \leq \cdots \leq \Delta_{(M)}$
- Initialize precision level : $\epsilon = 0$

Boundary Progression :

```
for m = 1 to M do
    if ϵ < Δ_(m) ≤ ϵ + c_T then
        Advance precision boundary : ϵ = Δ_(m) ;          // ϵ absorbs the gap
    else
        break ;                 // Indistinguishability band is empty
return ϵ* = ϵ
```

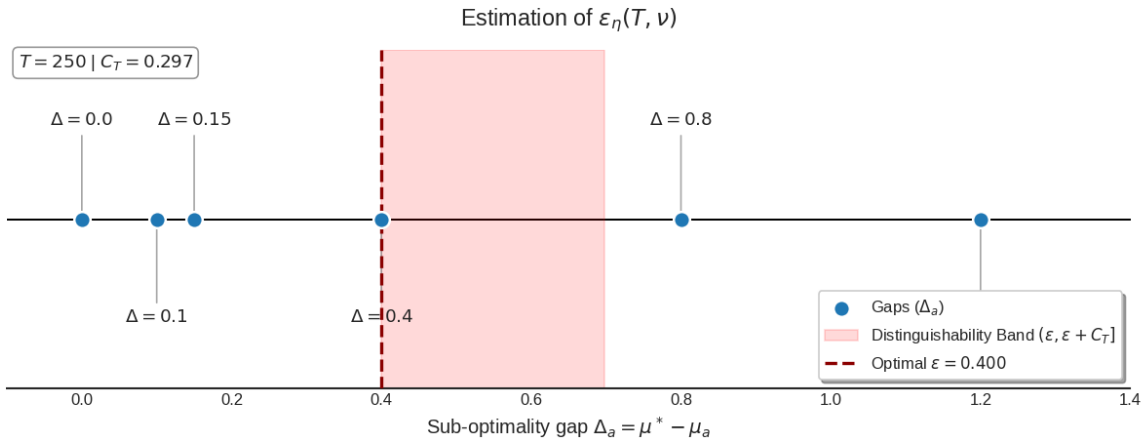

FIGURE 1.1 – $\epsilon_\eta(T,\nu)$ **in Gaussian environments.** Visualization of the sub-optimality space where arms Δ_a are partitioned into ϵ-optimal and statistically distinguishable sets. The red shaded area represents the distinguishability band $(\epsilon, \epsilon + C_T]$, defining the resolution limit of the practitioner for a given horizon T and efficiency $\eta = 0.5$.

1.3 $\epsilon_\eta(T, \nu)$ for Bernoulli Bandits (PAS FAIT POR L'INSTANT)

Consider a stochastic K-armed bandit environment ν. Each arm $a \in \mathcal{A}$ generates rewards according to a Bernoulli distribution $X_{a,t} \sim \text{Bernoulli}(\mu_a)$ where $\mu_a \in (0,1)$. We define the optimal mean reward as $\mu^* = \max_{a\in\mathcal{A}} \mu_a$.

For any precision level $\epsilon \geq 0$, recall the set of ϵ-near-optimal arms is $\mathcal{O}_\epsilon = \{a \in \mathcal{A} : \mu_a \geq \mu^* - \epsilon\}$.

1.3.1 Bernoulli Divergence

To compute the critical horizon $\tau_\eta^\epsilon(a)$ for a Bernoulli arm, we must evaluate the function $\mathcal{K}_{\text{inf}}(\nu_a, x) = \inf_{\nu:\mathbb{E}_\nu[X]\geq x} \text{KL}(\nu_a,\nu)$. When ν_a is Bernoulli with parameter μ_a, the distribution minimizing the divergence under the mean constraint is also Bernoulli. Consequently, for $x = \mu^* - \epsilon$:

— If $a \in \mathcal{O}_\epsilon$ (i.e., $\mu_a \geq \mu^* - \epsilon$), the constraint is already satisfied, so $\mathcal{K}_{\text{inf}}(\nu_a, \mu^* - \epsilon) = 0$. This implies $\tau_\eta^\epsilon(a) = 0$.

— If $a \notin \mathcal{O}_\epsilon$ (i.e., $\mu_a < \mu^* - \epsilon$), we have $\mathcal{K}_{\text{inf}}(\nu_a, \mu^* - \epsilon) = d(\mu_a, \mu^* - \epsilon)$, where $d(p,q) = p\log\left(\frac{p}{q}\right) + (1-p)\log\left(\frac{1-p}{1-q}\right)$.

For these sub-optimal arms, the condition $T \geq \tau_\eta^\epsilon(a)$ requires the divergence to strictly exceed the statistical threshold $\delta_T = \frac{\log T}{\eta T}$:

$$d(\mu_a, \mu^* - \epsilon) > \frac{\log T}{\eta T} \tag{1.15}$$

The practitioner definition requires $T \geq \max_a \tau_\eta^\epsilon(a)$. This means that, for a given horizon T and precision ϵ, every arm a must fall into one of two categories :

1. **ϵ-Optimal :** $\mu_a \geq \mu^* - \epsilon$.
2. **Statistically Distinguishable :** $\mu_a < \mu^* - \epsilon$ **and** $d(\mu_a, \mu^* - \epsilon) > \delta_T$.

An arm violates the practitioner condition if it is strictly sub-optimal at level ϵ, yet too close to be distinguished from $\mu^* - \epsilon$ at horizon T. The practitioner precision level $\epsilon_\eta(T,\nu)$ is the *infimum* $\epsilon \geq 0$ such that **no arm falls within this indistinguishability band** :

$$\epsilon_\eta(T,\nu) = \inf\left\{\epsilon \geq 0 : \forall a \notin \mathcal{O}_\epsilon,\ d(\mu_a, \mu^* - \epsilon) > \frac{\log T}{\eta T}\right\} \tag{1.16}$$

Estimating the practitioner precision level $\epsilon_\eta(T,\nu)$ requires finding the exact boundary of the indistinguishability band. For very short horizons (small T), the statistical threshold δ_T is extremely large. This pushes $\epsilon_\eta(T,\nu)$ to encompass the maximum sub-optimality gap ($\max \Delta_a$), meaning all arms fall into the ϵ-optimal set and the practitioner regret is exactly zero. As the horizon T increases, the statistical resolution improves, the indistinguishability band shrinks, and $\epsilon_\eta(T,\nu)$ gradually converges to zero, naturally recovering the classical asymptotic regret framework.

$$\epsilon_\eta(T,\nu) = \inf\left\{\epsilon \geq 0 : \forall a \text{ s.t. } \mu_a < \mu^* - \epsilon,\ \mu_a \log\left(\frac{\mu_a}{\mu^* - \epsilon}\right) + (1-\mu_a)\log\left(\frac{1-\mu_a}{1-(\mu^*-\epsilon)}\right) > \frac{\log T}{\eta T}\right\} \tag{1.17}$$

2 Lenient Regret with ε_t

2.1 Lenient Regret

Algorithm 2 ϵ_t-TS for Gaussian arms

Initialize $N_a(1) = 0$, $S_a(1) = 0$ and $\hat{\mu}_a(1) = 0$ for all $a \in [K]$ $\quad \triangleright$ Blue : Changes from TS
for $t = 1, \dots, T$ **do**
  for $a = 1 \dots, K$ **do**
    if $\hat{\mu}_a(t) > \max_j \hat{\mu}_j(t) - \epsilon_t$ **then**
      $\theta_a(t) = \hat{\mu}_a(t)$
    else
      $\mu_a(t) = \frac{\sigma^2\mu_0 + S_a(t)\sigma_0^2}{\sigma^2 + N_a(t)\sigma_0^2}$
      $\sigma_a^2(t) = \frac{\sigma^2\sigma_0^2}{\sigma^2 + N_a(t)\sigma_0^2}$
      $\theta_a(t) = \min\left(Y, \max_j \hat{\mu}_j(t) - \epsilon_t\right)$ for $Y \sim \mathcal{N}(\mu_a(t), \sigma_a^2(t))$
    end if
  end for
  Play $a_t \in \arg\max_a \theta_a(t)$ (break ties randomly) and observe the reward X_t
  Update $N_{a_t}(t+1) = N_{a_t}(t) + 1$, $S_{a_t}(t+1) = S_{a_t}(t) + X_t$ and $\hat{\mu}_{a_t}(t+1) = \frac{S_{a_t}(t+1)}{N_{a_t}(t+1)}$
  For all $a \neq a_t$, set $N_a(t+1) = N_a(t)$, $S_a(t) = S_a(t)$ and $\hat{\mu}_a(t+1) = \hat{\mu}_a(t)$
end for

Bibliographie